



Protocol Audit Report

Version 1.0

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Protocol Summary

PasswordStore is a protocol used to store and retrieve user passwords. It's designed for single-user use, not multi-user use. Only the owner can set and access passwords, and non-owners cannot.

Disclaimer

The Joev's team makes all effort to find as many vulnerabilities in the code in the given time period, but holds no responsibilities for the findings provided in this document. A security audit by the team is not an endorsement of the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the Solidity implementation of the contracts.

Risk Classification

		Impact		
		High	Medium	Low
Likelihood	High	H	H/M	M
	Medium	H/M	M	M/L
	Low	M	M/L	L

We use the CodeHawks severity matrix to determine severity. See the documentation for more details.

Audit Details

The findings described in this document correspond the following commit hash:

```
1 2e8f81e263b3a9d18fab4fb5c46805ffc10a9990
```

Scope

```
1 ./src/  
2 #-- PasswordStore.sol
```

Roles

- Owner: The user who can set the password and read the password.
- Outsiders: No one else should be able to set or read the password.

Executive Summary

As for the audit process, it didn't take much time because the protocol is simple and I didn't use any tools. In short, I found 3 vulnerabilities in the PasswordStore protocol, with 2 vulnerabilities rated HIGH and 1 vulnerabilities rated INFORMATIONAL. You can see and read the details below.

Issues found

Severity	Number of issues found
High	2
Medium	0
Low	0
Info	1
Total	3

Findings

High

[H-1] Storing the password on-chain makes it visible to anyone and no longer private

Description: All data stored on-chain is visible to anyone and can be read directly from the blockchain. The `PasswordStore : : s_password` variable is intended to be a private variable and only accessed through the `PasswordStore : : getPassword` function, which is intended to be only called by the owner of the contract.

We show one such method of reading any data off chain below.

Impact: Anyone can read the private password, severely breaking the functionality of the protocol.

Impact: Anyone can set/change the password of the contract, severely breaking the contract intended functionality.

Proof of Concept: Add the following to the `PasswordStore.t.sol` test file.

Code

```
1     function test_anyone_can_set_password(address randomAddress) public
2     {
3         vm.assume(randomAddress != owner);
4         vm.prank(randomAddress);
5         string memory expectedPassword = "myNewPassword";
6         passwordStore.setPassword(expectedPassword);
7
8         vm.prank(owner);
9         string memory actualPassword = passwordStore.getPassword();
10        assertEquals(actualPassword, expectedPassword);
11    }
```

Recommended Mitigation: Add an access control conditional to the `setPassword` function.

```
1  if(msg.sender != s_owner) {
2      revert PasswordStore__NotOwner();
3  }
```

Informational

[I-1] PasswordStore::getPassword natspec indicates a parameter that doesn't exist, causing the natspec to be incorrect.

Description:

```
1  /*
2   * @notice This allows only the owner to retrieve the password.
3   * @param newPassword The new password to set.
4   */
5  function getPassword() external view returns (string memory) {
```

The `PasswordStore::getPassword` function signature is `getPassword()` while the natspec says it should be `getPassword(string)`.

Impact: The natspec is incorrect.

Recommended Mitigation: Remove the incorrect natspec line.

```
1  -    * @param newPassword The new password to set.
```